

TANGUY, Russian Federation

WMO: 304050

Lat: 55.383N

Lon: 101.033E

Elev: 423

StdP: 96.35

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.5	-35.6	-40.7	0.1	-38.3	-37.8	0.1	-35.5	6.4	-10.8	5.4	-13.0	0.3	50	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	29.1	18.8	27.2	18.0	25.4	17.4	20.2	26.2	19.2	25.1	18.3	23.6	2.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.8	23.1	17.2	12.9	21.8	16.2	12.1	21.0	60.5	26.1	56.9	25.3	53.7	23.7	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.8	DB	-40.8	32.4	4.3	2.2	-43.9	34.0	-46.5	35.3	-48.9	36.6	-52.1	38.2	(4)
(5)			WB	-40.5	22.0	3.8	1.3	-43.2	22.9	-45.4	23.7	-47.6	24.4	-50.3	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-21.1	-17.7	-7.7	1.8	8.6	16.3	18.8	15.9	8.5	0.7	-10.2	-18.3	(6)
(7)		DBStd	15.00	7.95	7.86	6.80	4.65	4.69	4.14	2.86	3.62	4.15	5.01	7.61	8.51	(7)
(8)		HDD10.0	4482	966	774	550	248	85	4	0	3	78	290	607	877	(8)
(9)		HDD18.3	6879	1224	1008	808	496	303	87	29	90	296	548	857	1135	(9)
(10)		CDD10.0	729	0	0	0	2	42	191	273	187	33	1	0	0	(10)
(11)		CDD18.3	86	0	0	0	0	1	24	43	16	1	0	0	0	(11)
(12)		CDH23.3	880	0	0	0	0	34	307	361	174	4	0	0	0	(12)
(13)		CDH26.7	206	0	0	0	0	7	78	87	34	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	1.2	1.2	1.6	1.9	2.0	1.7	1.6	1.5	1.6	1.7	1.6	1.3	(14)
(15)	Precipitation	PrecAvg	377	14	11	10	16	33	60	62	68	36	23	22	18	(15)
(16)		PrecMax	538	20	23	19	51	61	143	107	134	82	50	45	45	(16)
(17)		PrecMin	208	6	4	2	3	12	18	9	21	4	6	8	9	(17)
(18)		PrecStd	96	4	5	5	11	15	32	27	30	18	12	9	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.3	1.0	9.0	17.8	27.1	31.6	31.7	29.9	23.8	14.6	6.5	-0.1	(19)
(20)			MCWB	-3.2	-2.1	3.1	8.4	14.3	18.9	19.8	19.4	16.0	8.6	2.9	-1.9	(20)
(21)		2%	DB	-5.2	-1.6	6.0	14.8	23.2	29.0	29.1	27.3	20.6	11.7	3.4	-3.1	(21)
(22)			MCWB	-6.2	-3.4	1.2	7.2	12.8	18.3	19.4	18.5	14.1	7.4	1.0	-4.1	(22)
(23)		5%	DB	-7.7	-4.2	3.6	12.0	20.2	27.0	27.1	25.2	17.7	9.4	1.4	-5.0	(23)
(24)			MCWB	-8.4	-5.6	-0.1	5.7	11.4	17.1	19.0	17.6	12.5	6.3	-0.3	-5.9	(24)
(25)		10%	DB	-10.3	-6.8	1.7	9.6	17.5	24.9	25.3	23.0	15.3	7.5	-0.2	-6.9	(25)
(26)			MCWB	-10.9	-7.9	-1.2	4.3	9.5	16.3	18.3	16.9	11.5	4.7	-1.5	-7.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.9	-1.1	3.6	9.5	14.9	21.4	22.0	20.7	16.7	9.9	3.4	-1.7	(27)
(28)			MCDWB	-2.2	0.7	8.4	16.2	25.5	28.0	28.4	27.2	22.6	13.1	5.6	-0.4	(28)
(29)		2%	WB	-6.1	-3.3	1.7	7.7	13.4	19.3	20.6	19.4	14.6	8.1	1.2	-4.0	(29)
(30)			MCDWB	-5.3	-1.6	5.3	13.2	21.4	26.4	26.3	25.7	19.0	10.8	2.9	-3.1	(30)
(31)		5%	WB	-8.4	-5.6	0.3	6.1	12.0	18.1	19.7	18.2	13.1	6.6	-0.1	-5.8	(31)
(32)			MCDWB	-7.7	-4.1	3.4	11.6	19.0	25.0	25.2	23.7	17.1	9.3	1.3	-5.1	(32)
(33)		10%	WB	-10.8	-7.9	-1.2	4.6	10.5	17.0	18.9	17.2	11.6	4.9	-1.6	-7.6	(33)
(34)			MCDWB	-10.3	-6.8	1.5	9.4	16.5	23.0	24.1	21.9	14.7	7.0	-0.1	-6.8	(34)
(35)	Mean Daily Temperature Range	MDBR		8.2	10.8	12.1	10.0	11.0	10.9	9.7	9.7	9.1	7.7	7.7	8.0	(35)
(36)		5% DB	MCDBR	8.3	11.3	13.3	14.2	16.0	15.4	13.7	14.5	13.5	11.6	7.8	7.5	(36)
(37)			MCWBR	7.7	9.5	9.3	8.3	7.7	6.2	5.7	6.6	7.3	7.4	6.1	7.0	(37)
(38)		5% WB	MCDBR	8.1	10.9	11.8	12.8	14.3	12.6	11.2	12.4	12.4	10.0	7.3	7.4	(38)
(39)			MCWBR	7.7	9.3	8.3	8.1	7.1	5.5	5.4	6.2	6.9	6.7	6.0	6.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.282	0.312	0.341	0.377	0.428	0.428	0.435	0.397	0.350	0.338	0.305	0.280	(40)
(41)		taud		2.015	2.003	2.023	2.190	2.116	2.221	2.226	2.316	2.407	2.278	2.115	1.980	(41)
(42)		Ebn at Noon		597	727	807	839	813	819	803	813	809	708	570	491	(42)
(43)		Edh at Noon		69	102	127	126	144	132	129	111	88	78	62	56	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.90	1.93	3.47	4.65	5.25	5.88	5.35	4.43	2.96	1.64	0.84	0.58	(44)
(45)		RadStd		0.07	0.09	0.17	0.34	0.48	0.42	0.45	0.28	0.32	0.16	0.09	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page